

3 Probability	3.1	Understand and use mutually exclusive and independent events when calculating probabilities. Link to discrete and continuous distributions.	Venn diagrams or tree diagrams may be used. Set notation to describe events may be used. Use of $P(B A) = P(B)$, $P(A B) = P(A)$, $P(A \cap B) = P(A) P(B)$ in connection with independent events. No formal knowledge of probability density functions is required but students should understand that area under the curve represents probability in the case of a continuous distribution.
	3.2	Understand and use conditional probability, including the use of tree diagrams, Venn diagrams, two-way tables. Understand and use the conditional probability formula $P(A B) = \frac{P(A \cap B)}{P(B)}$	Understanding and use of $P(A') = 1 - P(A),$ $P(A \cup B) = P(A) + P(B) - P(A \cap B),$ $P(A \cap B) = P(A) P(B A).$

Topics	What students need to learn:		
	Content		Guidance
3 Probability <i>continued</i>	3.3	Modelling with probability, including critiquing assumptions made and the likely effect of more realistic assumptions.	For example, questioning the assumption that a die or coin is fair.